

CSE 5524 Computer Vision - HW03 Implementation Explanations:

Question 6:

For Question 6, I made my own denoising filter by trial and error to try to clean up the noisy Dreese image without making it too blurry. The average and binomial filters both worked, but they made the image look too soft because they spread the weights evenly over a large area. To fix that, I made a 5 by 5 Gaussian-style kernels with higher weights in the middle, to try and keep the main details sharp while smoothing out high frequency noise areas. My implementation simply normalizes the kernel, so all values sum to one, then uses the convolution function I wrote earlier to apply it to the noisy image. The final image looked smoother overall but kept edges and lines clear, which I thought was a good middle ground.

Question 8:

For Question 8, I pulled one cosine pattern out of an image that mixed several together. I used a DFT, which turned the spatial patterns into points at specific frequency locations. Each cosine showed up as two bright points, one positive and one negative. Now having the target's frequencies, I created empty amplitude and phase lists and copied only those two corresponding cosine point values from the full DFT output. I set everything else to zero to filter out frequencies I had no need of. I passed the filtered amplitude and phase arrays into the IDFT function to reconstruct the target image. The final output matched the original cosine image almost exactly, which confirms that my implementation works as expected.